

LESSON 04 / 第四課

Pests, Pathogens and Predisposition: From Landscapes to Entry Courts

有害生物、病原與誘因：從地景到侵入門戶

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Biotic damage is not a random accident pasted onto an otherwise healthy stem. At landscape scale, host density and climate set outbreak envelopes. At tree scale, Manion's decline spiral still organises the literature: predisposing, inciting and contributing factors are different accounts. At organ scale the question is the entry court. Tubby and colleagues' review of published urban dieback case studies found that 81% named an arthropod or microorganism as the primary cause, while often discussing urban stress as predisposition. Demographic reviews, by contrast, rank stewardship and site. The two literatures sample different questions. This review keeps both in view and refuses a calendar spray without a threshold.

生物性傷害不是貼在原本健康莖上的隨機事故。地景尺度上，寄主密度與氣候設定爆發包絡。整株尺度上，Manion 的衰退螺旋仍組織文獻：誘因、激因與輔因是不同賬戶。器官尺度上，問題是侵入門戶。Tubby 及同事對已發表城市枯死案例的綜述發現，81% 以節肢動物或微生物為主因，同時常把城市逆境討論為誘因。人口學綜述則排序管護與立地。兩支文獻抽樣不同問題。本綜述兩者都保留，並拒絕沒有閾值的日曆噴藥。

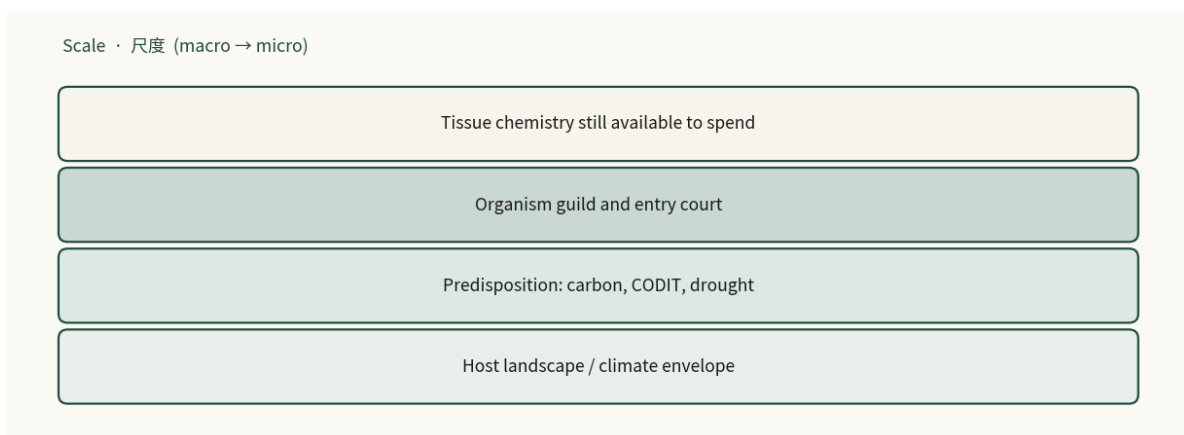


Figure 1. Nested scales from host landscapes to tissue entry courts.

圖1. 由寄主地景到組織侵入門戶的嵌套尺度。

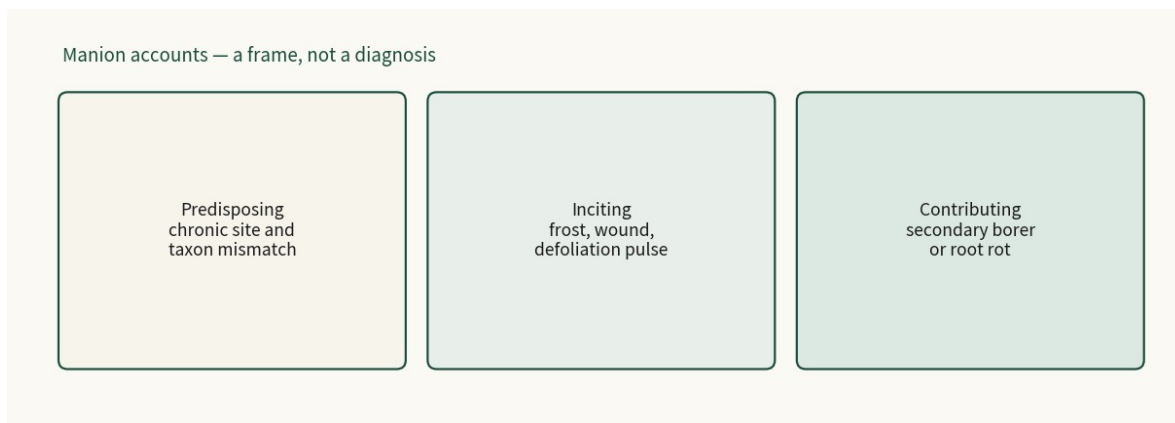


Figure 2. Manion's spiral as a bookkeeping frame, not a diagnosis.

圖2. Manion 螺旋作為簿記框架，不是診斷。

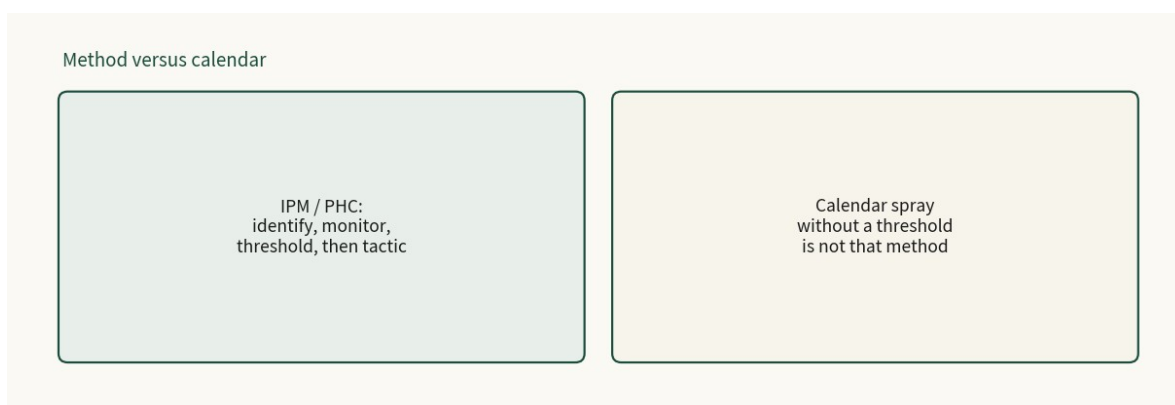


Figure 3. IPM/PHC with a threshold versus calendar treatment.

圖3. 帶閾值的IPM／植物健康照護，對比日曆處理。

1. Scope

1. 範圍

The object is published biotic damage and the predisposition that makes a host spendable. A regional pest list is an example, not the frame.

對象是已發表的生物性傷害，以及使寄主變得可被消耗的誘因。地區有害生物名單是例子，不是框架。

2. Landscape and predisposition

2. 地景與誘因

Host density, connectivity and climate envelope set whether a pest or pathogen can maintain an outbreak [1]. Monodominant urban planting lists are landscape experiments in host continuity; Santamour's diversity heuristic exists because that continuity has been expensive when a specialist arrives [2]. Manion's spiral separates predisposing factors (chronic: compaction, drought, inappropriate taxon), inciting factors (pulse: frost, defoliation, wound) and contributing factors (secondary colonisers) [3]. It is a bookkeeping frame. It does not name the organism on a given stem.

寄主密度、連通性與氣候包絡設定有害生物或病原能否維持爆發 [1]。單優的城市種植名單是寄主連續性的地景實驗；Santamour 的多樣性啟發式之所以存在，是因為當專性者到來時該連續性代價

昂貴 [2]。Manion 螺旋分開誘因（慢性：壓實、乾旱、不當分類群）、激因（脈衝：霜、落葉、傷口）與輔因（次生定殖者）[3]。它是簿記框架。它不為某一樹幹上的生物命名。

3. Two literatures

3. 兩支文獻

Tubby et al. found that case-study literature names biotic primaries more often than demographic reviews do; urban stress remains the usual predisposing discussion [4]. Hilbert-style demography, conversely, ranks taxon, size and stewardship [5]. Both can be true of different samples. Treating one literature as a refutation of the other confuses the sampling frame.

Tubby 等發現案例文獻比人口學綜述更常點名生物性主因；城市逆境仍是常見誘因討論 [4]。Hilbert 式人口學則排序分類群、大小與管護 [5]。兩者可以在不同樣本上同時為真。把一支文獻當成對另一支的反駁，是混淆了抽樣框。

4. Entry courts and method

4. 侵入門戶與方法

Root-rot basidiomycetes exploit living roots and buried wounds; sapwood borers exploit weakened cambium; foliar specialists exploit flush [6]. A fresh pruning cut is an entry court whose diameter is a dose. Integrated pest management and plant health care share a published order: identify, monitor, set a threshold, then choose the least-disruptive tactic that meets the objective [7]. A calendar spray without a threshold is not that method.

根腐擔子菌利用活根與埋藏傷口；邊材蛀蟲利用衰弱形成層；葉部專性者利用抽梢 [6]。新鮮修剪切面是侵入門戶，其直徑是劑量。整合有害生物管理與植物健康照護共享已發表順序：鑑定、監測、設閾值，然後選擇能達到目標且最少干擾的策略 [7]。沒有閾值的日曆噴藥不是該方法。

5. Limits

5. 限度

Diagnostic literature is richer in temperate pests than in tropical urban guilds. Molecular identification has changed names faster than management guides have been rewritten. The biology of the court usually survives the rename.

診斷文獻在溫帶有害生物上富於熱帶城市功能群。分子鑑定改名快於管理指南重寫。門戶的生物學通常在改名後仍在。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Manion's spiral is treated here as:

1. 此處把 Manion 螺旋視為：

A. A stem diagnosis.

A. 莖診斷。

B. A bookkeeping frame for three accounts.

B. 三本賬的簿記框架。

C. A t/R law.

C. t/R 定律。

D. A planting list.

D. 種植名單。

2. Tubby et al. found that published dieback case studies:

2. Tubby 等發現已發表枯死案例：

A. Never name biotic causes.

A. 從不點名生物性原因。

B. Named a biotic primary cause in 81% of papers.

B. 81% 論文以生物性為主因。

C. Proved soil pH is the only cause.

C. 證明土壤 pH 是唯一原因。

D. Replaced demography.

D. 取代人口學。

3. Demographic reviews most often rank:

3. 人口學綜述最常排序：

A. Only lightning.

A. 只有雷電。

B. Taxon, size/age, site and stewardship.

B. 分類群、大小／年齡、立地與管護。

C. Only P50.

C. 只有 P50。

D. Only Wall 4.

D. 只有第 4 壁。

4. A fresh pruning cut is:

4. 新鮮修剪切面是：

A. Never an entry court.

A. 絕不是侵入門戶。

B. An entry court whose diameter is a dose.

B. 直徑即劑量的侵入門戶。

C. A diversity rule.

C. 多樣性規則。

D. A flash map.

D. 閃擊圖。

5. IPM/PHC without a threshold is:

5. 沒有閾值的 IPM／PHC 是：

A. Complete.

A. 完整的。

B. Not the published method.

B. 不是已發表方法。

C. Required by CODIT.

C. CODIT 所要求。

D. Equal to 3-30-300.

D. 等於 3-30-300。

6. Monodominant planting lists are:

6. 單優種植名單是：

A. Irrelevant to outbreaks.

A. 與爆發無關。

B. Landscape experiments in host continuity.

B. 寄主連續性的地景實驗。

C. Hydraulic safety margins.

C. 水力安全邊際。

D. Barrier zones.

D. 屏障區。

7. Predisposing versus contributing factors are:

7. 誘因對輔因是：

A. The same account.

A. 同一賬戶。

B. Different accounts.

B. 不同賬戶。

C. Both t/R.

C. 都是 t/R。

D. Both inoculants.

D. 都是接種劑。

8. Root-rot fungi and foliar specialists:

8. 根腐菌與葉部專性者：

A. Share one entry court.

A. 共享一個侵入門戶。

B. Exploit different courts.

B. 利用不同門戶。

C. Are palms.

C. 是棕櫚。

D. Set SOS.

D. 設定 SOS。

9. Calendar spraying:

9. 日曆噴藥：

A. Is the definition of plant health care.

A. 是植物健康照護的定義。

B. Is not PHC unless a threshold is set.

B. 除非設閾值，否則不是 PHC。

C. Raises mycorrhizae.

C. 提高菌根。

D. Reprints section.

D. 重印截面。

10. Urban stress in the case-study literature is most often:

10. 案例文獻中的城市逆境最常是：

- A. Ignored.
- A. 被忽略。
- B. Discussed as predisposition.
- B. 被討論為誘因。
- C. Proof of zero biotic cause.
- C. 零生物性原因的證明。
- D. A cable grade.
- D. 鋼纜等級。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

- [1] Tubby K.V., Webber J.F. pest and pathogen outbreak envelopes in amenity trees.
- [2] Santamour F.S. Jr. (1990). Trees for urban planting: diversity, uniformity, and common sense. Metria 7.
- [3] Manion P.D. (1991). Tree Disease Concepts. 2nd ed.
- [4] Tubby K. et al. (2025). Systematic review of published urban tree dieback case studies.
- [5] Hilbert D.R. et al. (2019). Urban tree mortality: a literature review. AUF 45: 167–200.
- [6] Sinclair W.A., Lyon H.H. Diseases of Trees and Shrubs.
- [7] ISA Plant Health Care / IPM BMPs.
- [8] Edmonds R.L., Agee J.K., Gara R.I. forest health as predisposition plus agents.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。